

STATISTICS I

(CTS 125)

NDI PUBLIC ADMINISTRATION

Technical terminologies

Descriptive statistics: many experts look at descriptive statistics in different ways. Possible since there is no general accepted definition of the descriptive statistics. According to M. Usman et al (2011) viewed descriptive statistics as variables used to describe the basic features of data gathered from a field of study in various ways.

What these experts mean here is that descriptive statistics is a type of statistics that is used to explain the characteristics (attributes) of the analyzed data gathered or collected from field of study in different ways. Some experts also look at the descriptive statistics as one which can be interpreted through using of tables, charts or graphs.

However, looking at it scientifically, it is an art of collecting, organizing, summarizing, analyzing, presenting and interpreting of numerical data for better decision making processes in the face of uncertainty.

Inferential statistics: this consists of generalizing from a unit to some or more units. (i.e from sample to population). For example if a girl in a family displays good behavior outside her family, people tend to generalize such good behavior to the whole family members. The same is applicable when you use an instrument to fetch a small quantity of grains from a bag without Weevil's means the whole bag of the grains is without Weevils. Therefore, this type of statistics is used for testing, determining relationship among variables as well as making predictions or forecasting about analyzed sample data which represent the large population.

Class: this means group, category or sub-division of the total ranged values, e.g 10-15, 16-21, 22-27 etc.

Class limits: these are end numbers of each class e.g 10-15, the class limits of this group are 10 and 15.

Lower class limit: this is the smallest number in a particular class or group e.g 10-15, 16-21. The classes have 10 and 16 as their lower class limits.

Upper class limit: this is the largest or the highest number in a given class or group e.g 10-15, 16-21 classes, have 15 and 21 as their upper class limits since these numbers are the highest numbers in each of the classes or groups.

Class marks: this is the addition of lower and upper class limits of a class or group divide by 2.

Class size or width: this is the expected values or numbers in each class, e.g 20-26, the class size or width is 7, because we have values or numbers of 20, 21, 22, 23, 24, 25, and 26 in this group or class with 7 different values or numbers.

Range: this is the difference between the highest and the lowest values or numbers in a given set of data. For example, if we have the following values (age number of some children in a school), thus: 3, 4, 6, 7, 8 and 5. The range year is: $8-3=5$ years.

Midrange: this is addition of the lowest and highest number in a set of data divide by 2 e.g 1, 2, 3, 4 = $\frac{1+4+5+2}{2}=2.5$.

Variable: this is an attribute or characteristics that can assume different values e.g X can assume different values as either a, 1, girl, boy, man, woman, e.t.c. It is called variables because the value changes over time.

Parametric statistics: parametric statistics assume the type of statistics that a researcher uses samples gotten from population randomly to make inferences of the same population.

Non-parametric statistics: the non-parametric statistics assume the opposite of the parametric statistics whereby researcher does not take samples of population, but study the whole population individually to ascertain the results.

Statistics: its meaning, uses and importance

Business statistics is a familiar language or word which professionals and students use every day. Many scholars of statistics only attempted the definition of statistics without considering the business. But this text is unique in its way. And it may interest you to see that this text first looks at the definition of business and statistics differently before harnessing the two words together (business statistics).

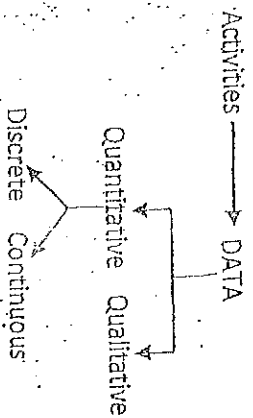
According to advance dictionary, business is the activity of making money by buying and selling of goods or provision of services, while statistics at the other hands, is a part of mathematics that deals with a set of numbers that represent facts, definite or measurement.

Looking at business statistics as a phrase is the scientific collection of numerical or non-numerical of economic activities (data) analysed necessary for better decision making.

To elucidate or make it clearer on numerical and non-numerical data are quantitative and qualitative variables. The quantitative variables represent or are the number values that can be ordered or ranked. For instance, the variable age is numerical as people can be ranked according to the value of their ages, number of oranges in a basket and other instances of quantitative variables are height, weights, as well as body temperatures. These quantitative variables are further divided into two. That's the discrete and continuous variables. Discrete variables are normally assigned values such as 0, 1, 2, 3 etc and can easily be counted. For instance, number of children in a family; 1, 2, 3, 4, 5, etc, number of students in a class; 1, 2, 3, 4, 5, 6, ----- 100, phone calls received from a loved one; 1, 2, 3 etc in a day.

Continuous variable: these are data that cannot be measured precisely of their values for instance, as earlier mentioned, temperature of a person is between 0°C to any number above 0°C ; i.e one's temperature could be 0°C - 20°C , 0°C - 30°C and 0°C - 37°C is always the normal temperature of human beings. Anyone with temperature below 0°C or above 37°C is abnormal. Heights are also measured using interval of points; from 0 inches to highest point say 7 inches (0 inches to 7 inches), (0 inches to 7.8 inches), etc. weights e.g from 0 gram to 3 grams, 0 gram to 5 grams, as children weights are measured in kilograms (kgs) when they are born in some hospitals. The non-numerical variables are those values that one can categorise, classify, or name without attributing any number to them. For instance a lecturer can announce to his or her students in the class that he or she would like to see all the male students in his or her office after the class. Here the lecturer is only referring to a category of gender (males students) only which does not involve any number that you can term as quantitative variable. Therefore we can see here that qualitative variables are all about classification, categorization, tagging, grouping of names in alphabetical order, classification of particular time are all known as qualitative variables.

Illustration of data variables



Data is normally classified as quantitative or qualitative variables, but these variables use measurement scales. These scales are of four types which are:

Nominal scale: this scale classifies data into mutually exclusive (non-overlapping) categories in which no order can be or ranking can be imposed on the data; for instance, we normally say maths teachers of this school or classification of male or female subjects (gender) into two groups as in male or female group.

Ordinal scale: this measurement or scale places subjects or data into categories and this in turn are placed in order or ranked as superior, average, or ordinary as the case may be. For instance, educational qualification certificate holders are not ranked the same in the labor market.

Interval scale or level of measurement: this normally differentiates between two units of measurement in existence. For instance, the measurement of a height of any individual differentiates the highest point of the height from the lowest point of the height, and this does not always mean zero (0) in the difference in such cases or measurement. Because from 0 inches to 7 inches, or from 0 inches to 3 inches if found the difference would not always mean 0 measurement.

The ratio scale of measurement: this measures two different members of a population, for instance, two different lecturers in an institution might have two different salaries which could be termed as ratio variable and also becomes ordinal variable.

Illustration of different measurement scales

Nominal level of measurement(data)	Ordinal level of measurement(data)	Interval level of measurement(data)	Ratio level of measurement(data)
Gender (male or female)	Grade (A, B, C)	IQ	Height
Political parties (PDP, APC, etc)	Rating scale (poor, average, good, very good)	Temperature	Weight
Religion groups	Judging-guilty, innocence.	Weight	Time
Nationality			Salaries
Nigerians etc.			Age

What are data? Data in statistics are seen as unorganized number, letters, symbols or objects e.Lc that are needed to be processed into meaningful information, while information is the processed data that are explicitly meaningful in nature. Note that data can be used as singular or plural.

Uses of business statistics

The various ways that business statistics can be used are as follows:

- It is used to know the existing condition of a variable say a business setting.
- It is used to answer a long range planning questions of when and where to locate business or facilities in an environment.
- It is used for decision making and forecasting in business or organization.
- It is used to guide business research.
- It is used to measure progress of a variable in a place.
- It is used to find out the number of people in a country (census).

Note: statistics is used in all fields of endeavors, be it medicine, education, geology, energy e.t.c and its effects cannot be overemphasized.

Importance of statistical data

It is very important here to alert readers or users of this text the difference between the uses and the importance of statistical data so as not to be confused altogether. Use of objects or beings are roles, functions, purpose, utility, consumption or habit that those objects or beings can perform, while importance is the greatness, splendidence, immenseness, enormouness, magnificence or distinguishness those objects or beings possess.

It is worthy of note that statistical data is not an object or being that can be touched and felt, but a subject which is an idea, a field of study, a discipline, case, topic, issue, subject, matter, or theme. Therefore the importance or significance of statistical data are as follows:

- a) Presentation of facts in a definite form: this means making conclusions in a clearer way numerically is better than conclusion stated qualitatively, for example, it's always better and clearer to make a statement that we sold 200, 000 liters of fuel yesterday than saying we sold a lot of fuel yesterday.
- b) Statistics reduce complexities of data into intelligent summary at a glance. For instance, some voluminous data can be reduced to totals, average, percentages e.t.c at a glance.
- c) Statistics classify numerical facts: human beings are full of mistakes and errors hence cannot group a multitude of figures; for instance the group of certain scores in a grade level of (0-39)f, (40-45)D, (46-49)DC, (50-55)C, (56-59)CB, (60-65)B, (66-69)BA, (70 and above)A, e.t.c help us to form accurate and more precise idea about the performance of the class of students.
- d) It is a good tool for comparison of facts: certain facts may be meaningless if not compared with similar facts at the other periods of primary and secondary time, for instance the national income of a country does not possess value itself until it is compared with the past one to know whether the standard of living is on the increase (improved living) or decrease (poor living) standard.

Sources and methods of data collection

There are two main sources of data which are Primary and secondary sources of data. Primary sources of data are undocumented, first hands sources that researchers use for the purpose of the study. For example, personal interview, designed questionnaire, telephone interview, group street interview and direct observation.

Secondary sources of data are those sources-origins-documented or already made information that the researcher lays his/her hands (access) for the purpose of study. For example, employment Gazette, textbooks, newspapers, journals and articles, magazines, recorded videos or tape.

Below are detail discussions on both primary and secondary sources of data.

Primary sources

- a) Personal interview: this is the process of asking the respondent face to face about the subject matter of the study.
- b) Designed questionnaire: this is a set of written questions to the respondent to provide answers to the questions asked.
- c) Telephone interview: it's a process whereby a researcher calls the respondent (interviewee) on phone purposely for the study on ground.
- d) Group street interview: at times, the researchers meet groups of people on the streets, and find out issues concerning the study at hand.
- e) Direct observation: this is the personal watching of events, subjects or being in order to establish facts about the events, subjects or the being.

Secondary sources of data

- a) Employment gazette: this is a summary of employment record concerning the employees like date of appointment, rank, sex, age, nationality, nature of responsibility or task to assume, salary scale etc for a period of time.
- b) Text-books: these are books written by experts or scholars for contact of information by researchers or whoever is in need of them.
- c) Newspapers: these are daily publications of different information by different organizations of such papers like punch, New Nigeria, Sun, Tribune etc.
- d) Journals and articles: these are also publications that are mainly for professionals e.g Journal for school of business administrations and management, Journal for medical practitioners, Journal for mechanical and electrical engineering etc.
- e) Magazines: these are spasmodic (periodic) publications containing stories and articles for enlightenment of information needed for research.

Problems of data collection

Some of the problems a researcher experiences or encounters in the course of collecting data are considered below:

- a) Financial constraints: most at times a researcher would like to obtain some information relevant to his/her study, but would not be backed up financially to set such information. For example, there could be some information in the net that is highly relevant for one's study, but the amount of foreign exchange attached to it is unaffordable by the researcher.
- b) Language barrier: some information that need to be obtained easily becomes so hard because the researcher and the respondent cannot understand themselves or each other.
- c) Illiteracy: when a researcher cannot read or write fluently, there is bound to be problem in the cause of collecting the data.

- d) Lack of patience: some researchers do not have enough patience (heart) to wait for their respondents to provide needed answers to the questions asked hence end up in wrong study or research.
- e) Lack of materials: some researchers don't have access to the materials need to carry out the study intended. For instance, there are some machines that detect diseases, but unavailability of them make many patients to die ignorantly.

How to Design Questionnaire

The topic of one's research determines the design of questionnaire. For instance, if the topic reads like this "influence of customers as side effect on some provision stores in Mubi North L.G.A Adamawa state". Any researchable topic must contain two variables which are dependent and independent variables. From the topic, the dependent variable are the provision stores in Mubi North L.G.A Adamawa state- which is also known as resultant or outcome variables while the independent variables is the influence of customers as side effect which is also known as explanatory variables that can be manipulated by the researchers.

Provision stores in Mubi North L.G.A Adamawa state depend on the attitudes of customers that patronize or buy goods from them which is a reason of their dependent variables. In other words, the topic can be looked at as setback of some provision stores in Mubi North L.G.A Adamawa state as caused by their customers. Now for one to write a set of written questions (questionnaire) that would guide one to provide solution to the topic, one must first think of the purpose (objective) that one would like to achieve at the end of the research or the study.

In this case, one among other objectives that the researcher would like to achieve is to find out or discover the facts or problems that caused the setback of the provision stores by the customers. Therefore all the questions that ought to be asked should be geared towards the finding out of the causes of the setback of the stores so as to proffer solution to them by way of recommendations after analysis of the answers/data provided to the questions asked.

Types of questionnaires: there are two types of questionnaires which are open and close ended questionnaires. The open ended questionnaires (unstructured) are those types that a researcher does not provide options or answers to the questions asked while the close ended questionnaires is that type that the researcher provides options of answers to the questions asked for the respondents to choose their choice of answers by either ticking (✓), or cross (×).

The necessary steps for the formation of questionnaires

Firstly, letter of introduction: here the researcher writes letter to his/her respondents identifying his/herself either as student researcher or otherwise and the topic of the study.

In the second paragraph of the letter the researcher uses polite words to tell the respondents how the questionnaires can be answered, and finally ensuring the respondents the confidentiality of the information to be provided. For example:

Department of Business Admin &
Management,
Federal Polytechnic Mubi,
Adamawa state.
30th June, 2017.

Dear respondent,
Request to fill questionnaire
I am a final year student with the above address undertaking a research titled "influence of customers as side effect on some provision stores in Mubi North L.G.A Adamawa state. You are kindly requested to answer the attached questionnaire by ticking against the appropriate answers. All information provided shall remain highly confidential for this study only.
Thanks in anticipation for your cooperation.

Yours faithfully,
Tizhe Medugu
SBT/Bam/HND/5/001

Secondly, the questions are attached to the above letter: the questions here are normally divided into two parts or sections. The first section contains the personal data of the respondents, while the second section contains the questions to be answered by respondents. For example, using the topic "influence of customers as side effect on some provision stores in Mubi North L.G.A Adamawa state" could be thus:

Questionnaire

Section A: personal data of the respondent.

Sex: male ☐ female ☐

Marital status: married ☐ single ☐

Age: 15-20 ☐ 21-25 ☐ 26-30 ☐

Educational qualification: ND ☐ NCE ☐ Degree ☐

Section B: please tick the appropriate answers or options that best correspond to your views (opinions).

1. The owner of provision store A has been convicted of selling fake goods many times.
A: strongly agree ☐ B: agree ☐ C: disagree ☐ D: strongly disagree ☐
2. The price items of this provision store A is relatively higher than others.
A: strongly agree ☐ B: agree ☐ C: disagree ☐ D: strongly disagree ☐
3. The owner of this provision store A has no customer relations.
A: strongly agree ☐ B: agree ☐ C: disagree ☐ D: strongly disagree ☐
4. People always found their stolen goods in store A.
A: strongly agree ☐ B: agree ☐ C: disagree ☐ D: strongly disagree ☐
5. The owner of provision store A does not give credit to customers like others.
A: strongly agree ☐ B: agree ☐ C: disagree ☐ D: strongly disagree ☐

The above questionnaire is a brief sample of how to design questionnaire which after collection of the answers from the respondents would be analyzed using the appropriate statistical tool which we shall discuss in later chapter.

Errors committed while collecting data

- a) Wrong questionnaire: if the questions written to the respondents to answer are wrong, then the data that would be collected will definitely be wrong also.
- b) Wrong instrument: if a faulty instrument is used for collection of data, definitely there shall be bound to be error in that data collection, for example, if a doctor uses wrong thermometer to collect body temperature of a patient, there would be error of in that date collection too.
- c) Lack of understanding the questionnaire: if respondents do not understand the questions asked would give wrong answers which invariably would lead to wrong data collection.
- d) Carelessness and forgetfulness: sometimes, copies of questionnaire shared out to the respondents are not collected back by the researchers. This results as uncompleted copies.

Also, at times, information collected verbally as in interview with the respondents can be forgotten if the data are voluminous.

Population and basic sampling procedures

In statistics, a population refers to the totality of set of people, animals, or objects that are found in a place over a period of time. A population is finite or sizeable if researcher can mail every member of such population, while the infinite population is that one that the researcher cannot reach out to all the people individually e.g large market or the students in a particular institution.

Since a researcher hardly can reach out to all members of infinite population, we now talk of a definite (sample) population. Sample of population is therefore a part of the infinite population randomly selected by a researcher for study. For instance, a buyer of bag of beans in market normally fetches a small portion from the bags each as sample to know the best among the bags so as to generalize conclusion before buying.

The little quantity of beans fetched by the buyer indeed possess all the characteristics of the unfetched ones in the bags. This is exactly what we do in research. We randomly select people, animals or objects out of many that cannot be reached individually, study the randomly selected ones (sample) and refers whatever found out (characteristics) to the major group that was not studied to be the same of the studied ones.

Advantages of sample study

- a) It saves time: sample saves time by not studying all the members of the infinite population, since the sample possess characteristics of the major population unstudied.
- b) It serves as reference: whatever a researcher finds out from the sample study is generalized to the infinite population not studied.
- c) It saves cost: a huge amount that would have been spent for studying infinite population is now reduced by studying small population (sample).
- d) It creates relationship between the researcher and the respondents: the researchers easily reach the members because they are small in number (part of infinite population), hence, creates relationship between them.

Probability and non-probability samplings

The process or the techniques adopted for selection of sample is called probability sampling. Some of the probability samplings are as follows:

- a) Random sampling: this is method of sampling where every members of the population has equal right to be selected from the population for the sample to be studied. If sample is selected from a population randomly, it is called unbiased sampling, but if otherwise i.e. if it is not selected randomly, it is called bias selection of sample. In that case, the results of the study or research would be invalid or unreliable.

The procedures for simple random samplings are:

(a) Ballot papers: this is a situation where a researcher writes on pieces of papers yes for those that would be selected, and no for those that would not be selected for the study. For example, if a researcher has a population of 2,000 students and wants a sample of 200 students for the study, they would then write yes on 200 pieces of paper, and no on 1800 pieces of paper and mix them together. The researcher then would ask the 2000 students to start picking the pieces of paper one after the other, and those that pick yes of the 200 pieces of the papers become the sample size of the population for the study. In this way, the researcher is not biased in the selection of the population sample size for the study.

(b) Table or otherwise known as scientific method of sampling: This is a situation where a researcher pre-arranges the whole population of the study in columns and rows in a table depending on the required columns and rows in alternation. The researcher is then expected to choose a number and proceed the selection vertically or horizontally bearing in mind the rules of ignoring any number more than the total population size and any number less than the first chosen number selected because they are not within assigned sample members produce the required sample for the study.

Example 1.7: suppose a study has a population of two groups in Gombe Market (men and women) each 25 bringing the total to fifty (50), and from which ten (10) of them are to be selected as sample with the following instructions:

- I. Formation of a table with (5) columns and ten (10) rows in alternation of male and female.
- II. Select from column two 32 to proceed vertically and produce the designed sample (10 members) bearing in mind the rules guiding table random selection of sample.

Solution

- 1) Formation of pre-arrangement of scientific table sampling of 1-50 population with five (5) columns and ten (10) rows in alternation of male and female or female or female and male respectively is thus:

Table formation:

C1	C2	C3	C4	C5
R ₁ 1 m	2f	3 m	4 f	5 m
R ₂ 6 f	7 m	8 f	9 m	10 f
R ₃ 11 m	12 f	13 m	14 f	15 m
R ₄ 16 f	17 m	18 f	19 m	20 f
R ₅ 21 m	22 f	23 m	24 f	25 m
R ₆ 26 f	27 m	28 f	29 m	30 f
R ₇ 31 m	32 f	33 m	34 f	35 m
R ₈ 36 f	37 m	38 f	39 m	40 f
R ₉ 41 m	42 f	43 m	44 f	45 m
R ₁₀ 46 f	47 m	48 f	49 m	50 f

- 2) From the table and in column 2, 32 is selected and proceeded vertically to produce the required or deserved sample thus bearing in mind the rules guiding table random selection of sample, we have 32f, 37m, 42f, 47m, 33m, 38f, 43m, 48f, 34f and 39m as the ten members of sample for the study.

Note: other questions can be generated from table like what number and sex is the interception of C3R67 E.t.c.

(b) **Systematic sampling:** this is otherwise known as quasi random type of sampling or selection because its population is homogeneous in nature. It is quasi and systematic because once the first number is selected randomly, the subsequent numbers will be automatically selected.

Example 1.8: let us assume there is a population (N) which its sample size (n) has to be selected from using this formula; $N=nK$, where N =population size, n =sample size and K an integer, that n can divide both sides so as to get the value of K .

Use systematic (random sampling) to select 8 members as sample size (n) from a population of 80 (N) given that the first number selected randomly is 5.

Procedures

1. Assign numbers 1-80 randomly to members of the population.
2. Select randomly a number x such that $1 < x < K$.
3. Select the numbers $x, x+K, x+2K, \dots, x+(n-1)K$.
4. The population members with the assigned numbers form the sample.

Solution

Using the formula $N=nK$, and the variables into the formula, we have

$$80=8K=80/8=8K/8 \therefore K=10.$$

Step 1 assign numbers 1-80 randomly to members of the population.

Step 2 select randomly a number x such that $1 < x < K$, and our $x=5$ and $K=10$

Step 3 select the numbers, $x, x+K, x+2K, x+3K, x+4K, x+5K, x+6K, x+7K, x+(n-1)K$ the 8th term = (75).

These numbers are 5, 5+10, 5+20, 5+30, 5+40, 5+50, 5+60, 5+70= 5, 15, 25, 35, 45, 55, 65, and 75

form the numbers of the 8 people as the sample (n) size that are selected from the population 80 (N).

Confirm the 8th term using this formula given $x+(n-1)K=75$, $\therefore 5+(8-1)10=5+(7)10=5+70=75$.

(c) **Stratified random sampling:** this is another method also that is found among homogeneous population (similar characteristics), for example If a population of one hundred (100) people consists of 30 rich class and 70 poor class; and a sample of 20 people are to be selected from the rich and the poor classes to make up for the sample size of the population.

Procedures

- Take the ratios of the two classes using the population numbers multiplied by the total number of sample (n) size to know the sample number of each class.
- Add the two sample numbers gotten together and this forms the total sample (n) size (20) of the population for the study.

Solution

The ratio of the poor class $\frac{70}{100} \times \frac{20}{1} = 14$,

The ratio of the rich class $\frac{30}{100} \times \frac{20}{1} = 6$.

Sample size of the poor class is 14.

Sample size of the rich class is 6.

Therefore, the total sample size (n) is 20.

(d) Cluster sampling: this is when a population is vast (large) and is spread across geographical areas. In this type of sampling, the population is first being shared into clusters and units for the sample to be selected. For instance, if a study is to be carried out on a whole state e.g Adamawa state.

Procedures

1. Divide the state into regions which in turn can be divided into wards or villages. Sample can therefore be selected from these units for study.

Non-probability sampling: we have already defined non-probability sampling as a process of selecting samples where the individuals does not have equal rights to be selected for study.

Type of non-probability samplings

- Accidental convenience sampling: this is simply the situation where researchers demand for volunteers of those who are willing to participate in the study. For instance, those who are willing to bath corpses before they are buried.
- Purposive sampling: this is a situation where a researcher has a pre-determined group to serve the purpose of the study, for instance, the study of matrilocalists (men being married by women).
- Expert sampling: these are professional people being selected for study, e.g accountants, doctors, lawyers, engineers, teachers, etc.

Methods of data presentation and analysis

Data are usually presented in form of charts, tables, depiction of pictures and even graphs to find over-all interpretation of a situation at hand. Some of these charts are as follows:

Bar-chart: this is a diagrammatic representation of statistical data constituting of individual or separated bars of equal widths whose heights are drawn proportional to the frequencies of the classes they belong or represented.

Example 2.1

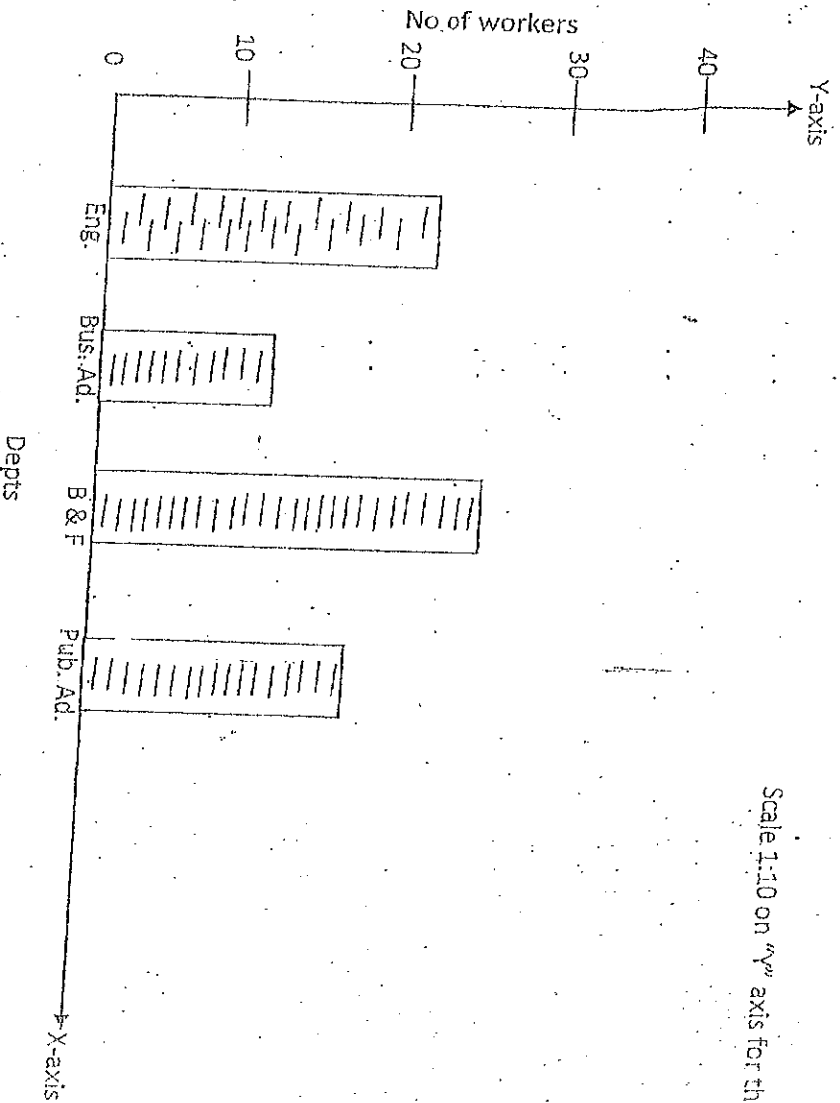
The table below shows some departments with different cadre of workers in an institution.

Departments	Junior	Middle	Senior	Ex.	Total
Engineering	10	6	5	2	23
Business Admin	6	3	2	1	12
Banking & finance	12	8	4	3	27
Public Admin	6	5	4	2	17
Total	34	22	15	8	79

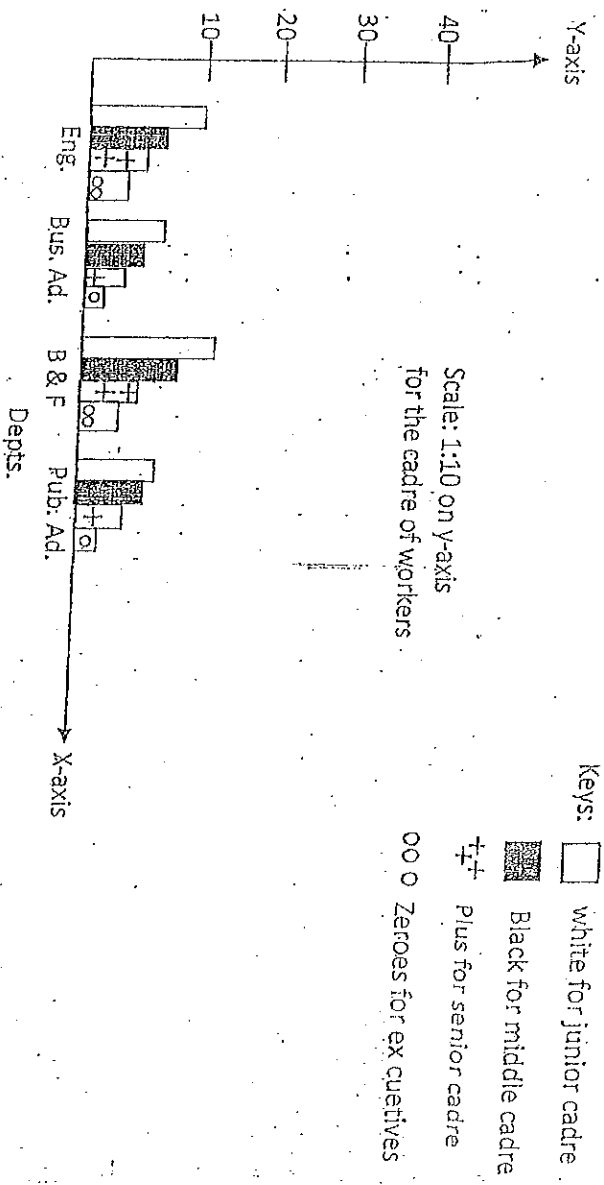
This table shows two way frequencies horizontally as all the departments with different cadre of workers each with their totals. The overall total of the horizontal frequencies is always equal to the overall total of vertical frequencies as shown above seventy nine (79) each respectively.

To present the above data in a simple bar chart, we make use of the total number of workers in each department only on "y" axis without referring to the worker's cadres or classes and the departments on "x" axis. The total workers in each of the departments are 23, 12, 27 and 17 under a graph knowing the variables that are for "y" and "x" axes. These variables to be noted are the total workers in each department and the departments respectively. Finally, take note of these totals to know the appropriate scale that can always be used. Imagine the appropriate scale that can fit the graph by looking at the number of workers in each department.

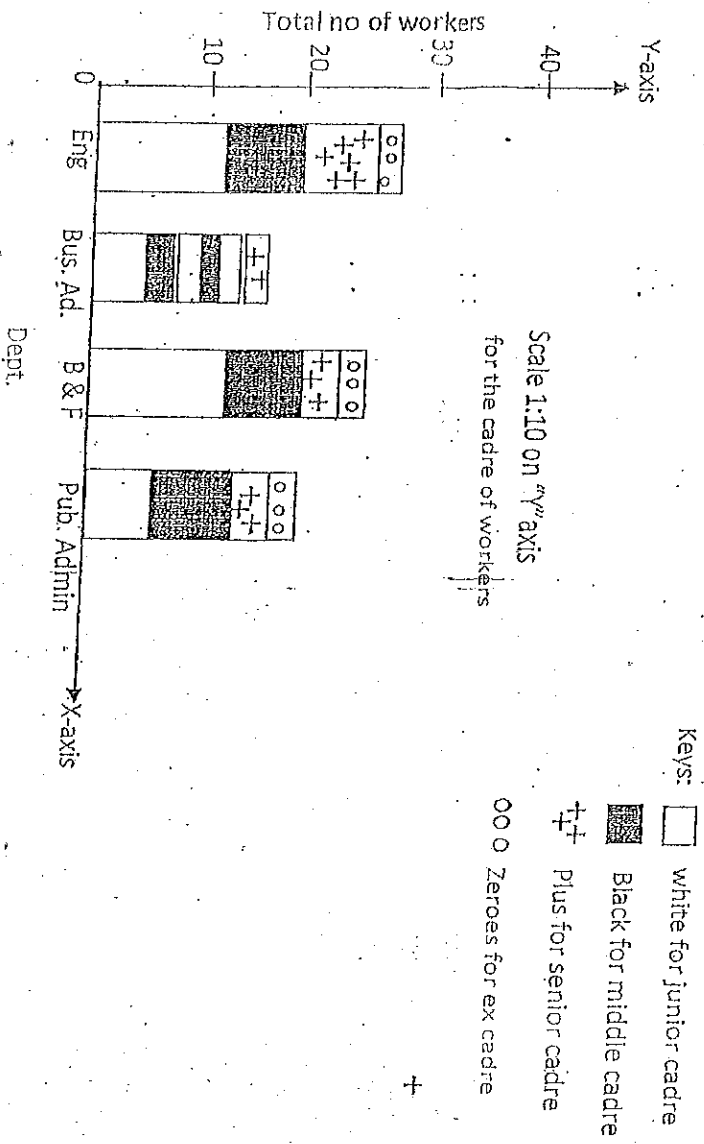
Scale 1:10 on "Y" axis for the workers



Multiple bar chart: this is when two or more variables are considered together in a single bar or department as in the case of our example. To plot this chart on a graph, each of these cadres or cluster of workers above or as given in the table in each department has to be marked or given a special sign each as a key given to the different cadre of workers. Therefore below are different keys given to the different cadres in the departments.



Component bar chart: this is a situation where two or more variables are presented in a single bar.
Each variable (cadre) or class is segmented by indicating its key.



In a situation where the variables are to be expressed in percentage form, each bar will possess 100% composition divided by the total variable. For instance, to get the percentage of junior cadre in the department of engineering is as this:

$$\text{Junior} = \frac{10}{23} \times \frac{100}{1} = 43.4\% \approx 43\%.$$

$$\text{Middle} = \frac{6}{23} \times \frac{100}{1} = 26.1\% \approx 26\%.$$

$$\text{Senior} = \frac{5}{23} \times \frac{100}{1} = 21.7\% \approx 22\%.$$

$$\text{Executive} = \frac{2}{23} \times \frac{100}{1} = 8.7 = 09\%.$$

100%

If the percentage calculated above are summed up and the total is below or above 100%, then there is a problem in the calculation re-check your work.

Pie chart: this is a circular diagram which represents 360° statistically. The circle is divided into segments and each segment is proportional to the degree of the variable being represented in the circle chart. For instance, to have the pie chart representation of the different departments being engineering, Business administration, banking and finance and public administration, we take each department total workers divided by the total grand of all the workers in the departments multiply by 360 as done below:

$$\text{Engineering department} = \frac{23}{79} \times \frac{360}{1} = 104.8^\circ$$

$$\text{Business Admin department} = \frac{12}{79} \times \frac{360}{1} = 54.7^\circ$$

$$\text{Banking and finance department} = \frac{27}{79} \times \frac{360}{1} = 123.0^\circ$$

$$\text{Public Admin department} = \frac{17}{79} \times \frac{360}{1} = 77.5^\circ$$

$$\text{Total} \longrightarrow 360.0^\circ$$

To present the degrees got in pie chart, use a pair of compasses, protractor, pencil and a ruler to construct the pie chart indicating the different departments in their various degrees.

Procedures

- Fix your pencil tightly to the hollow part of the pair of compasses.
- Place the pointed part of the pair of compasses on the center of your protractor and carefully turn your hand to either left or right hand side to construct a circle.
- Take a radius of the circle with your ruler and pencil. Place your protractor on the radius to measure the depts. in degrees as calculated and apportion them accordingly.

Histogram: this is usually a continuous information on a joined vertical or horizontal bars (meeting at class boundaries). What makes this, different from the bar chart is that bar chart has its bars separated while histogram has its bars joined together continuously.

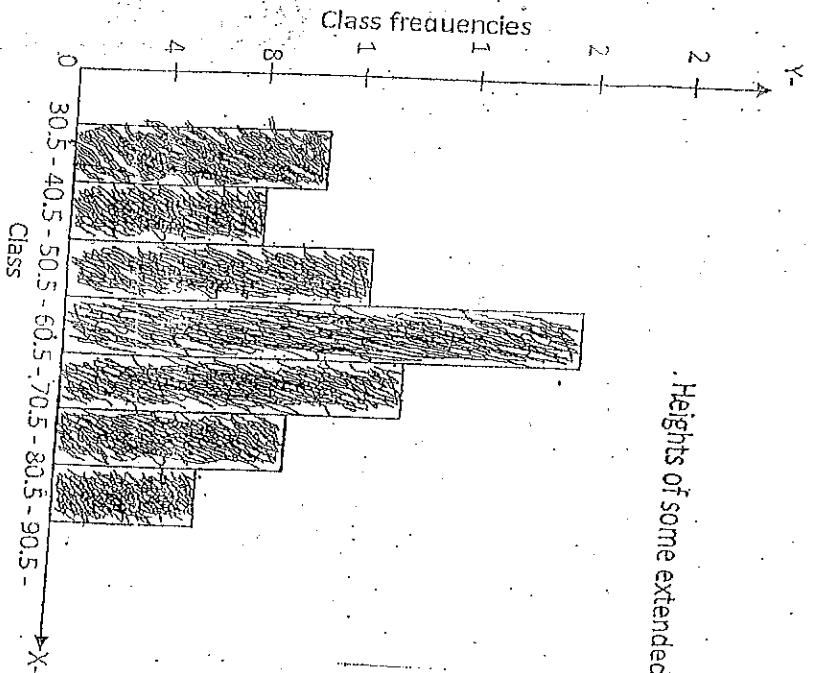
Example 22: below are some groups of extended family members' heights. Use the data to construct a histogram.

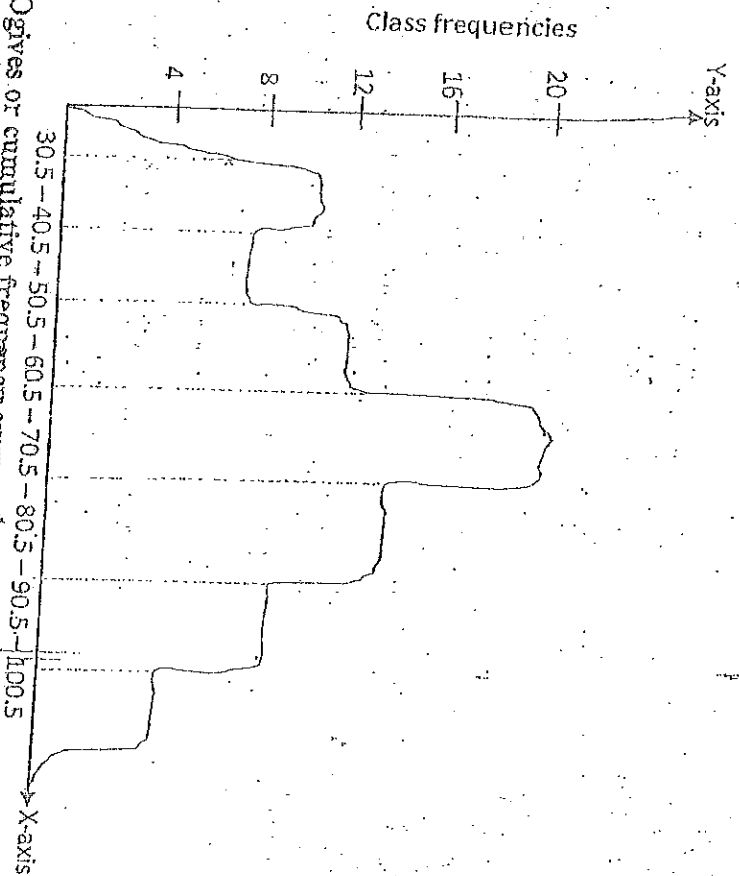
Class Limits	Class Boundaries	Class Frequencies
31-40	30.5-40.5	11
41-50	40.5-50.5	8
51-60	50.5-60.5	13
61-70	60.5-70.5	19
71-80	70.5-80.5	14
81-90	80.5-90.5	9
91-100	90.5-100.5	6

Now to represent the above data on a graph as histogram, choose a scale by looking at the frequencies bearing in mind the lowest frequency of the data using 1:4 scale on "y" axis (frequencies) and on "x" axis boundaries, we can easily plot our histogram vertically on a graph.

Frequency polygon: this is a graphic representation of histogram in a curve form as shown below

Heights of some extended family





O gives or cumulative frequency distribution. It is a situation whereby a number of items in a distribution whose values are more than a certain amount are represented on a graph. In this case, if one is interested to know the number of items that are "less than" others in a certain group, the accumulation will proceed from the least to the greatest size in the distribution, and the numbers so obtained or gotten will be called "less than" cumulative frequency distribution, and if one desires to know more than a certain size, accumulation will proceed from the greatest to the least and the series is called more than cumulative distribution.

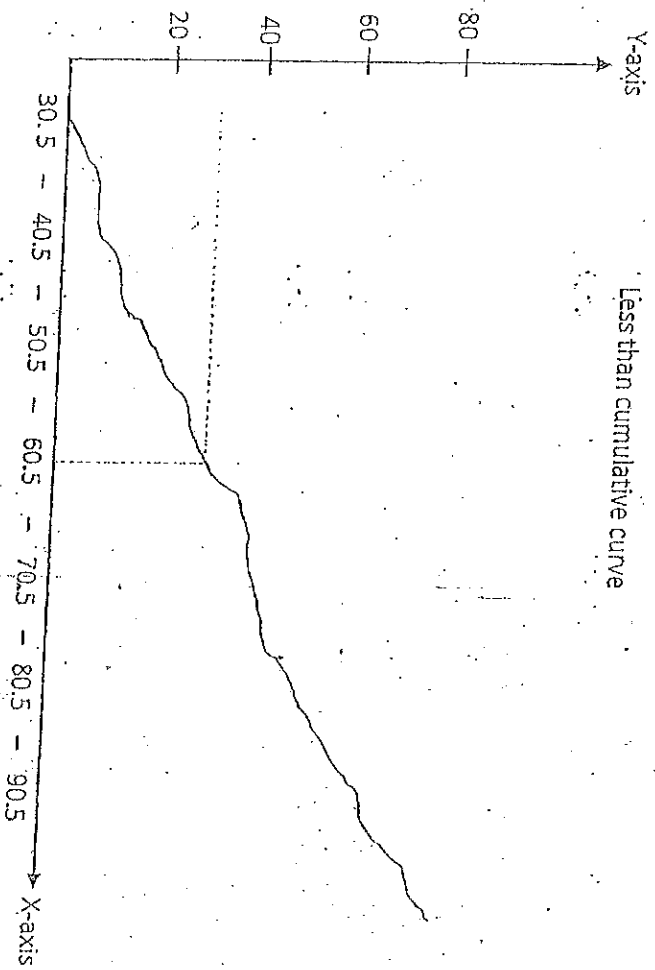
Example 23: using the extended family members' heights, in example 22 table, the construction of a less than cumulative frequency distribution is presented thus:

Less than cumulated frequency table

Class Boundary	Cumulative Frequency
30.5 or less	11
40.5 or less	19
50.5 or less	32
60.5 or less	51
70.5 or less	65
80.5 or less	74

90.5 or less	80
--------------	----

With the help of this table, it is easier therefore to find the number of the extended family members whose heights are equal or less than a certain heights thus there are 74 extended family members whose heights are 80.5 inches or less.

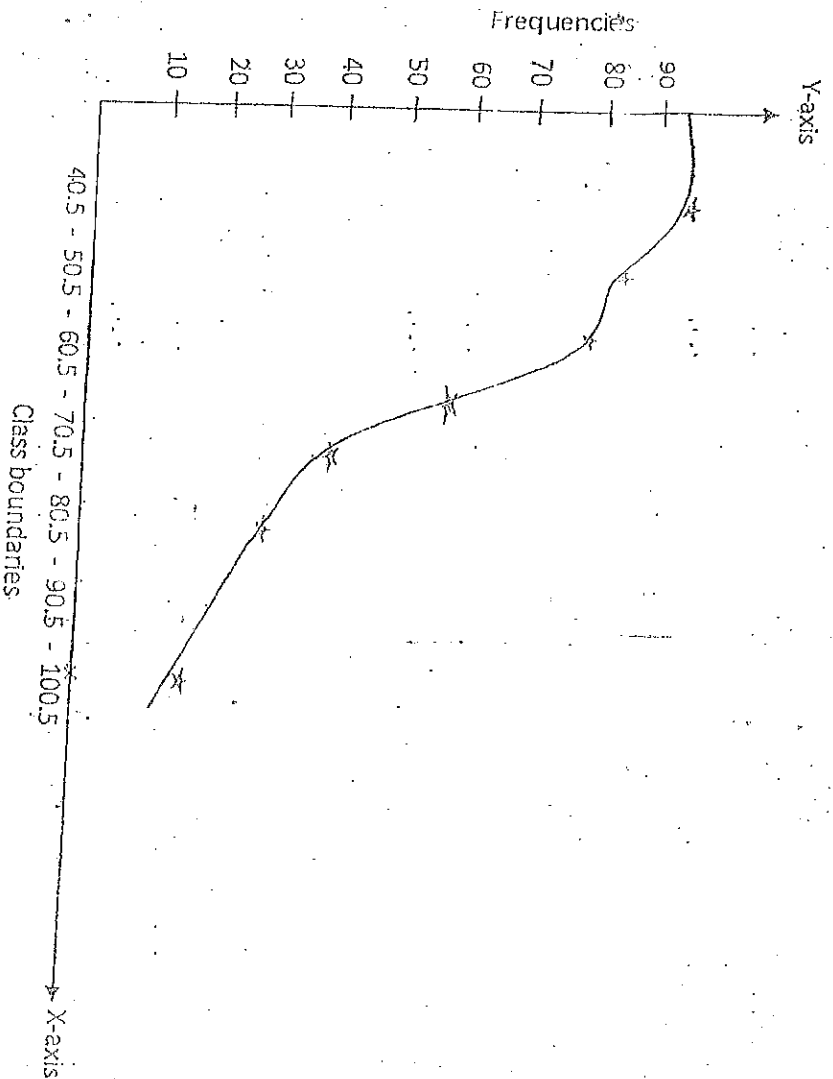


Usually, a "more than" cumulative frequency table is constructed starting with the upper end of the distribution. Using our example 22, the extended family members whose heights are 90.5 inches or more are $6+0=6$. Also, the total number of family members whose heights are 70.5 or more are $29+15+6=40$.

More than cumulative frequency table

Class boundary	Cumulative frequency
40.5 or more	80
50.5 or more	69
60.5 or more	61
70.5 or more	29
80.5 or more	15
90.5 or more	6
100.5 or more	0

When this table is presented or is plotted on a graph, the curve so obtained is known as an Ogive pronounced as *Ojive*.



Exercises

- I. What are the necessary methods of presenting and analyzing descriptive set of data?
- II. What is the difference between the plotting method of simple bar chart and that of multiple bar chart?
- III. How can a variable be represented in a pie chart?
- IV. What do you understand by Ogive and how is it pronounced?
- V. What is the main objective of plotting Ogive on a graph?

Measures of central tendencies are the cardinal variables that statisticians could obtain from a set of data. These cardinal variables are range, mean, median and mode. It is also known as

measures of central location. This means assessment of fundamental, primal or key elements in a set of data. Range is the difference between the highest and the lowest values in a set of data.

Mean which is also known as average is the center of the distribution or the most typical case. Again it is the number that all the members of a distribution have it equally. The median is the

middle number of a set of items when arranged in order of magnitude while mode is the most frequented item in a set of data. Other specialized averages are quadratic, harmonic and geometric means.

Important notes: measures found by using all the data values in the population are called parameters, while measures obtained by using data values of samples are called statistics. Therefore an average obtained from a sample of data representative is called a statistic and the average obtained from the entire population is called parameter.

The mean which is also known as the arithmetic average is calculated by adding all the values of the data and dividing it by the total number of values of observations.

Example 3.1, the monthly income of some security guards with the federal Polytechnic Nnobi are as follows: ₦12, 000, ₦18, 000, ₦20, 000, ₦30, 000, ₦16, 000, ₦25, 000, ₦22, 000, ₦15, 000. To find the mean salary of the guards, we use the formula of ungrouped data or sample mean which is $\bar{x} = \frac{\sum x}{n}$, where (Sigma) stands for summation or addition, $\bar{x}$ stands for salaries, and n for no. of the total security guards.

Procedures

- I. Arrange the data in order of magnitude and in addition.
- II. Sum-up the salaries and divide it by the total number of the guards (n).

Therefore, the total number of guards (n):

$$\bar{x} = \frac{\sum x}{n} = \frac{₦12,000+₦15,000+₦16,000+₦18,000+₦20,000+₦22,000+₦25,000+₦30,000}{8} = \frac{₦158,000}{8} = ₦19,750.$$

This means if the different monthly income (variations) are combined together and shared among the eight (8) security guards equally, each of them would get ₦19,750.00. It is equal shares of items among people in a community or a society. When you hear an average Nigerian eats 2 times a day means if all foods are gathered together and share among Nigerians each Nigerian would eat twice in a day.

The range salary from the example is therefore the highest salary value which is ₦30,000 minus the lowest salary value ₦12,000 equal to ₦18,000.

Arithmetic mean ($\bar{x}$) for grouped data

Formula= $\bar{x} = \frac{\sum fx}{\sum f}$; where $\bar{x}$ is the symbol for sample mean, $\sum$ is the summation symbol (Sigma),

f is the frequency of the data or observations, x is the midpoint or class mark of each class interval. fx is the frequency representing the total number of the given observations i.e $\sum f$ is the sum of the frequency representing the total number of the given observations i.e $\sum f = n$.

Example 3.4: below are the scores obtained by some students in a test of statistics 212.

Scores

Class interval	Frequency or No. of students
21-30	4
31-40	6
41-50	13
51-60	15
61-70	9
71-80	3

Required to calculate the mean score of the test. Show your workings.

Procedures: always remember the formula to use in any statistical question given. Therefore, the formula for grouped mean is:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

i.

Find the class mark or midpoint (x).

Here we add the lower and upper of each class interval and divide by 2 thus:

$$21-30 = 21+30 = 51 \div 2 = 25.5$$

$$31-40 = 31+40 = 71 \div 2 = 35.5$$

$$41-50 = 41+50 = 91 \div 2 = 45.5$$

$$51-60 = 51+60 = 111 \div 2 = 55.5$$

$$61-70 = 61+70 = 131 \div 2 = 65.5$$

$$71-80 = 71+80 = 151 \div 2 = 75.5$$

Multiply your final answers above as your class members or midpoints by their various frequencies.

$$fx = 4 \times 25.5 = 102.$$

$$6 \times 35.5 = 213.$$

$$13 \times 45.5 = 591.5$$

$$15 \times 55.5 = 832.5$$

$$9 \times 65.5 = 589.5$$

$$3 \times 75.5 = 226.5$$

Therefore, $\sum fx = 2,555.0$. Also, insert these figures in fx column as below.

Scores/class interval	Frequency	Midpoint/class mark	fx
21-30	4	25.5	102
31-40	6	35.5	213
41-50	13	45.5	591.5
61-70	9	65.5	832.5
71-80	3	75.5	226.5
Total	$\sum f = 50$		$\sum fx = 2555$

$$\text{Therefore, } \bar{x} = \frac{\sum fx}{\sum f} = \frac{2555}{50} = 51.5.$$

This means the mean ($\bar{x}$) score of the test in statistics 212 is 51.5.

For a population, the Greek letter μ (mu) is used for mean, $\mu = \frac{X_1 + X_2 + X_3 + \dots + X_N}{N} = 51.5$

Where N represents the total number of values in the population.

Simple frequency distribution: this consists of a list of data values each showing the number of items having that value called frequency.

Example 3.5: the following simple frequency distribution resulted from examining the time some students come for statistics class in a whole semester.

No of days late for class-	0	1	2	3	4	5	6	7	8
No of students (frequency)	41	52	18	11	5	2	4	0	1

Required to provide:

- The day that the majority of the students were late to the class during the semester.
- Students that did not late to the class at all during the semester.
- The most common number of days that the students were late to school during the semester.
- What is the range of days late for class?

Example 3.6: the following data show the number of days some students used to travel down from their various states in Nigeria to school for 1st and 3rd semester.

1	1	2	3	4	0	2	2	1	1
0	0	1	2	3	4	5	2	2	0
3	3	2	1	1	1	0	2	4	2
0	1	2	2	1	3	3	4	4	0
5	4	2	0	0	1	2	3	4	1

Required to construct simple frequency distribution using tally chart.
How to form grouped frequency distribution i.e how to form class-intervals from a set of distribution.

Procedures

1. Calculate the range of the set of data, i.e identify the smallest and the biggest numbers and subtract the smallest number from the biggest to give you the answer as the range of the data distribution.
2. Number of classes or class-intervals is: $1 + 3.322 \log N$ where N is the sample size. (extremes) i.e the numbers that are too far from the numbers found in the distribution, they can be taken account of by making the first and/or last class open-ended.
4. The frequency table distribution can now be formed using a tally chart.

Example: the table below shows a number of orders received by a company each week over a period of fifty weeks. Compile or form a grouped frequency distribution

24	13	28	15	25	29	15	46
9	10	17	22	23	17	16	32
11	12	18	20	13	27	18	22
20	14	26	14	19	19	40	31
17	21	23	26	18	24	21	27

Solution

1st step: lowest value = 9, highest value = 46, Range = 46 - 9 = 37.
2nd step: $N = \text{Number of classes or class intervals} = 1 + 3.322 \log N = 1 + 3.322 \log 50 = 6.43 = 7$
3rd step: class width or size = $37/7 = 5.28$ and this is adjusted upward to 6.

3rd step: the formation of the classes are as seen below with the 1st class having the value of the lowest value, nine (9) and last class with the highest value, forty-six (46) in the distribution.

4th step: use the tallies to form the frequency.

No of orders	Tallies	Frequencies
5-9	I	1
10-14	IIII II	7
15-19	IIII IIII I	11
20-24	IIII II	10
25-29	II	2
30-34		0
35-39		1
40-44		0
45-49		1
Total		40

Another method that you can adopt to form grouped frequency

1. Determine the range i.e $H - L$ values equal to "range" class intervals by this formula i.e $1 + \frac{3.322 \log N}{\log 2}$ where N is the number of the sample size.
2. Determine class span, size or width by dividing the range by the number of class intervals

Example: use the same number of orders received by a company each week to compile or compute grouped frequency distribution using the second method.

Solution

1. The highest value in the distribution is 46, while the lowest value in the distribution is 9. Therefore, the range is equal to $46 - 9 = 37$.
2. To determine the number of classes or class intervals by formula is $1 + \frac{3.322 \log N}{\log 2}$ where N is the number of sample size which is 40. Therefore, $1 + \frac{3.322 \times \log 40}{\log 2} = 1 + \frac{3.322 \times 1.602059991}{1} = 6.322043291$
3. Class span, width or size = $\text{Range} \div \text{number of class intervals}$ got in 2. Therefore, $37/6 = 6.166666667$

Note: the lowest number in the set must be found in the first class, and the highest in the last class. We can now form the group frequency table.

Class interval	Tallies	Frequency
5-10	II	2
11-16	III III	10
17-22	III III III	13
23-28	III III III	10
29-34	I	3
35-40	I	1
41-46	I	1
Total of N as sample size = 40		1

Sigma notation: Sigma is a symbol in Greek written as Σ which means "sum of" in English and stands for the same in Latin. Let $X_1, X_2, \dots, X_n$ be a set of n observations or measurement, the $\sum_{i=1}^n x_i$ where $i=1, 2, \dots, n$. For example, if $X_1 = 33, X_2 = 20, X_3 = 31, X_4 = 25$ and $X_5 = 52$. Required to represent these scores of five students in statistics 212 in Greek Sigma notation.

Solution

The sum of these scores of five students in statistics 212 in Greek Sigma notation.

$$\sum_{i=1}^5 x_i = x_1 + x_2 + x_3 + x_4 + x_5 = 33 + 20 + 31 + 25 + 52 = 161.$$

The operational rules of Σ (Sigma): the rules that this Σ (Sigma) obeys are easily applied than its basic definitions.

$$\sum_{i=1}^N c \times x_i = c \sum_{i=1}^N x_i \text{ where } c \text{ is constant}$$

For example, if the scores $x_i = 5, 7, 9, 11$, and 13 are selected randomly from NDI purchasing and supply department, and the same students scored the same marks in another course. Applying the above formula in Greek, we have:

$$\begin{aligned} \sum_{i=1}^5 2x_i \text{ (where } i = 5, 7, 9, 11 \text{ and } 13) &= (2 \times 5) + (2 \times 7) + (2 \times 9) + (2 \times 11) + (2 \times 13) \\ &= 10 + 14 + 18 + 22 + 26 = 90. \end{aligned}$$

Or

$$2(x_1 + x_2 + x_3 + x_4 + x_5) = 2(5 + 7 + 9 + 11 + 13) = 2(45) = 90$$

Therefore, $\sum_{i=1}^N c_i = Nc_i$ where c_i is a constant.

Example, if seven staff are selected @ random from Business Admin department Federal Polytechnic Mubi and each is having the same number of students to lecture ninety (90). The total Greek is written as thus:

$$\sum_{i=1}^7 c_i = Nc_i \text{ where } c_i \text{ is a constant } \therefore 7 \times 90 = 630 \text{ where } c_i = 90.$$

The median: this is an item that lies exactly half way along a set of distribution arranged in order of magnitude. When a set of data contains an even number of items, there is no unique middle or center value. What happens in this situation is that we take average of the two middle items to give us the practical (median) item.

When calculating the median of a set of data, rank the N observation which stands for the total numbers of the observations. If N is odd, the median is the value of the observation with rank $N+1/2$.

Example 3.7: Below shows a number of times each student in a group visited library in a particular month: 15, 10, 8, 20, 13, 17 and 9 times. Find the median.

Procedures

1. Arrange the data in order of magnitude. Therefore, 8, 9, 10, 13, 15, 17, 20. Since the total number of observations (N) is odd, that's $N = 7$. The median is the observation and the rank $\frac{7+1}{2} = \frac{8}{2} = 4$ the 4th rank; and the fourth rank in order of magnitude is 13 which is the median.
2. If the N observation is an even number, then the median is the average value of the two middle observations that lie half way between the whole observations N with ranks $N/2$ and $\frac{N}{2} + 1$.

Example 3.8: Below are ten (10) public administration students that go into the library with different sizes of bathroom slippers: 10, 16, 15, 8, 9, 6, 11, 13, 7 and 12. Find the median size of their slippers.

Procedures

Arrange the set of data in order of magnitude as shown below:
6, 7, 8, 9, 10, 11, 12, 13, 15 and 16.

Using our even number of observation (N) which is ten here with its formula as $\frac{N}{2} + 1$, where our $N = 10$ th rank, we have $\frac{N}{2} = \frac{10}{2} = 5$ th rank, and $\frac{N}{2} + 1 = \frac{10}{2} + 1 = 5 + 1 = 6$ th rank. $\therefore$ the 5th and 6th ranks in the observations are 10 and 11, we then have the median as $\frac{10+11}{2} = \frac{21}{2} = 10.5$

How to find median from a grouped data

For a grouped data, the median is calculated using this formula:

$$\text{Median } (M) = L + \frac{\left\{\frac{N}{2} - c_f\right\}}{f_m} \times c, \text{ Where}$$

L : is the lower class boundary of the medium class.

N : is the total number of the observations.

f_m : is the frequency of the median class.

c_f : is the cumulative frequency up to but not including the median class.

c : is the class span or size of the class interval or width.

Example 3.9: Below are marks of one hundred (100) students of Business Admin department who sat for BAM 222 examination.

Scores/class interval	Frequency
11 - 20	4
21 - 30	8
31 - 40	16
41 - 50	26
51 - 60	24
61 - 70	14
71 - 80	5
81 - 90	3
	$\sum f = 100$

Solution and procedures

1.

From class boundaries and cumulative frequency columns added to the 1st columns by subtracting point five (.5) and adding point five (.5) of the lower and upper rule in the class interval each respectively. Place them each on the class boundaries column as can be seen.

Class intervals/scores	Class boundaries	Frequency	Cumulative frequency
11 - 20	10.5 - 20.5	4	4
21 - 30	20.5 - 30.5	8	12
31 - 40	30.5 - 40.5	16	28
41 - 50	40.5 - 50.5	26	54
51 - 60	50.5 - 60.5	24	78
61 - 70	60.5 - 70.5	14	92
71 - 80	70.5 - 80.5	5	97
81 - 90	80.5 - 90.5	3	100

2nd step: find the value of $\frac{N}{2}$ or $N \div 2$ where our $N = \sum f$

$$\therefore \frac{N}{2} = \frac{100}{2} = 50.$$

3rd step: find the cumulative frequency (cf) value that first exceeds $\frac{N}{2}$ which identifies the median class. The cumulative frequency value (cf) that first exceeds $\frac{N}{2}$ (50) which identifies the median class is 54.

4th step: having known the median class, we can substitute our variables into the formula thus:

$$\text{Median} = 40.5 + \left\{ \frac{\frac{100}{2} - 28}{26} \right\} \times 10$$

$$\text{Median} = 40.5 + \left\{ \frac{50 - 28}{26} \right\} \times 10$$

$$\text{Median} = 40.5 + \left\{ \frac{22}{26} \right\} \times 10$$

$$\text{Median} = 40.5 + \frac{220}{26}$$

$$\text{Median} = 40.5 + 8.4615$$

Median = 48.9615 $\approx$ 49 marks.

Estimation of median graphically

A percentage cumulative frequency curve or Ogive pronounced as Ojive is drawn and the value of the variable that corresponds to 50% point (i.e. halfway along the distribution) is read off and given the median estimate.

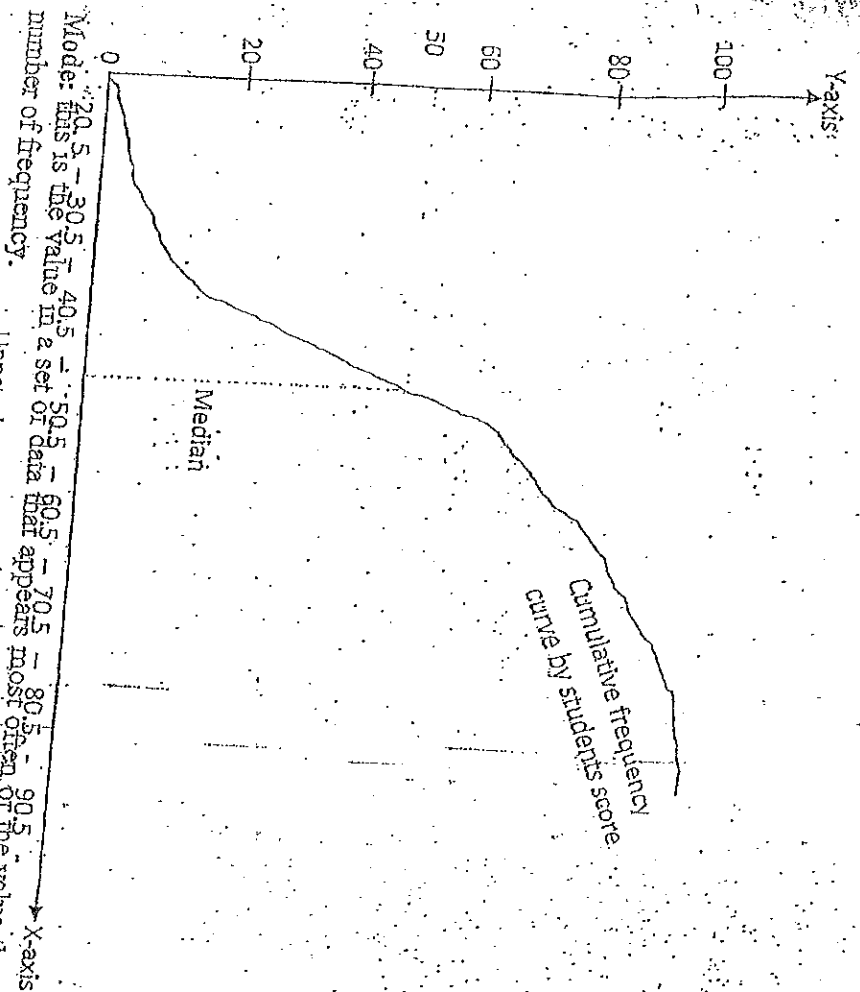
Procedures

1. Form three (3) columns for upper class boundaries, cumulative frequency percentage (cf %) and record all the upper class boundaries in its column, also that of cumulative frequency (cf) and
2. Take percentage of each cumulative frequency (cf) and place the results each on the (cf %) column.
3. Draw up a cumulative frequency curve by plotting class upper of upper class boundaries against cumulative percentage frequency and join the points with a smooth curve. Read off the 50% point to find the median.

Using our example 3.6, the procedures are as follows thus:

1. Formation of the three columns i.e upper class boundaries only, cumulative frequency (cf) and the cumulative frequency percentage (cf %).

Upper class boundary	Cumulative frequencies	Cumulative frequencies percentage (cf %)
20.5	4	4%
30.5	12	12%
40.5	28	28%
50.5	54	54%
60.5	78	78%
70.5	92	92%
80.5	97	97%
90.5	100	100%



Example: The following are scores obtained by some students.

Scores-	5	6	7	8	9	10
Frequency-	5	21	18	9	2	

Required to identify the mode. The mode of this simple discrete frequency distribution is 6 since this value has the largest frequency or the one that appears most often or the one that has the highest number in the distribution. The above is in terms of ungrouped data, but when it is a grouped data the mode is gotten through the following procedures using this formula thus:

$$Mode = L + \left\{ \frac{D_1}{D_1 + D_2} \right\} \times C, \text{ where:}$$

- i. Determine the modal class by that class which has the largest frequency
- ii. Calculate D_1 = the difference between the largest frequency immediately preceding (before) it
- iii. Calculate D_2 = the difference between the largest frequency (f) and the frequency immediately following (after) it.
- iv. C = stands for width class. The formula is thus:

$$\text{Mode} = L + \left\{ \frac{D_1}{D_1 + D_2} \right\} \times C, \text{ where:}$$

L = lower boundary of modal class.

C = modal class width.

D_1 = the difference between the largest frequency (f) and the frequency immediately preceding (before) it.

D_2 = the difference between the largest frequency (f) and the frequency immediately preceding (after) it.

Example 3.11: The following are the ages of some students in Business Admin. Department of Federal Polytechnic Mubi.

Age (years):	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44
No of students:	2	14	29	43	33

Required to find the modal age of the students.

Solution

The formula for calculating mode for grouped data should first be stated thus:

$$\text{Mode} = L + \left\{ \frac{D_1}{D_1 + D_2} \right\} \times C, \text{ where } L \text{ is the lower class boundary of the modal class, } D_1 \text{ is the difference between the largest frequency (} f \text{) and the frequency immediately preceding (before) it. } D_2 \text{ is the difference between the largest frequency (} f \text{) and the frequency immediately after it. } C \text{ is the width of the class interval.}$$

Step 1: determination of the modal class through table presentation of the data.

Age (years)	No of students
20 - 24	2
25 - 29	14
30 - 34	29
35 - 39	43
40 - 44	33

The modal class here is 35 - 39 since it is the class that has the largest frequency (43).

$D_1 = 14$, since it is the difference between the largest frequency and the frequency immediately preceding (before) it.

$D_2 = 10$ since it is the difference between the largest frequency and the frequency immediately following (after) it.

$C =$ the class width is 5 because from 35 years to 39 years are 5.
Having known all the variables, we substitute them into the formula to arrive at the final answer.

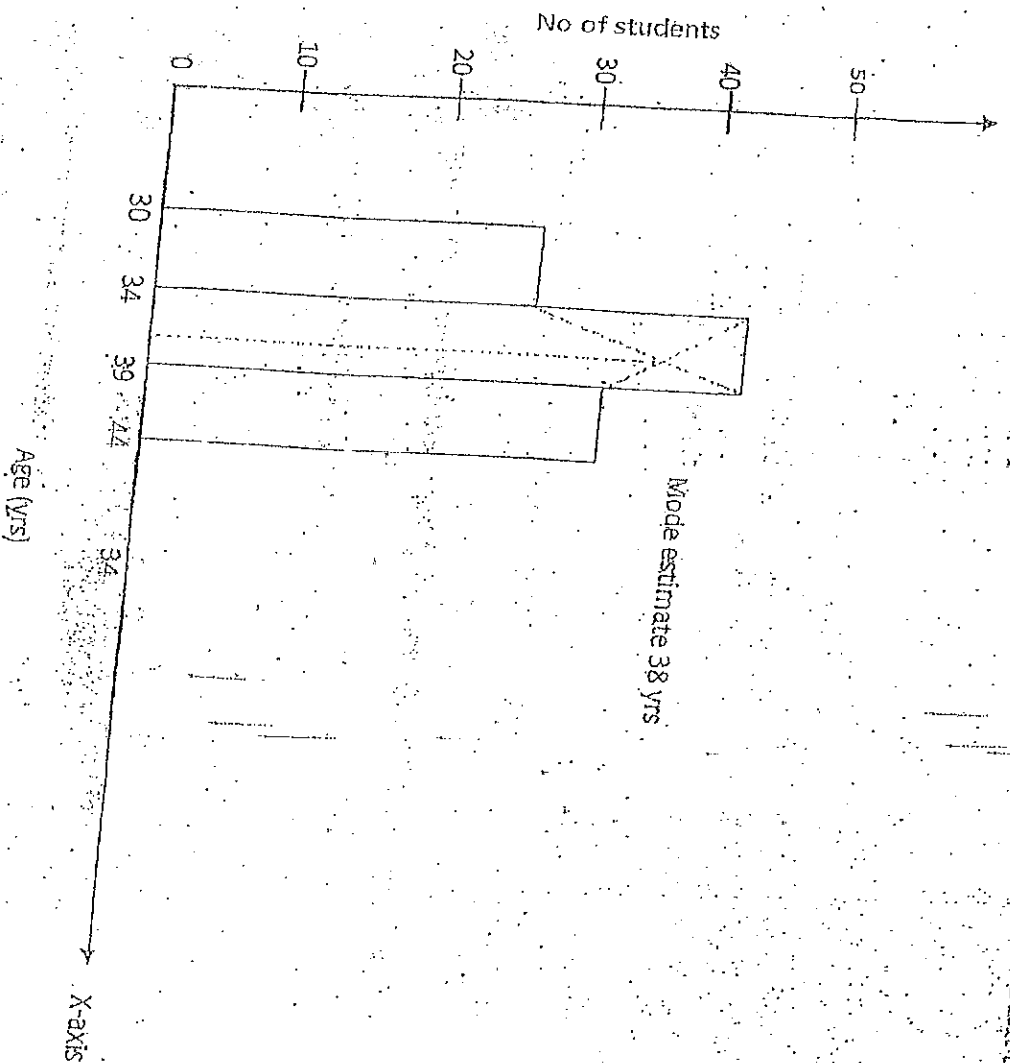
$$\text{Mode} = 35 + \left\{ \frac{14}{14+10} \right\} 5 = 35 + \left\{ \frac{14}{24} \right\} 5$$

$$= 35 + \{0.58\}5 = 35 + 0.58 \times 5$$

$$= 35 + 2.916 = 37.9 \approx 38 \text{ years.}$$

Graphical Representation of Mode Procedures

Simply construct three (3) histogram bars on a graph with the highest number of frequency in the middle and the ones on either side of it. Draw two lines from the vertexes of the bar that has the highest frequency to meet the other two vertexes attached on the either side from left to right.



and right to left respectively. Draw a straight line from the intersection of the two lines which has the highest frequency bar to X-axis line to determine the mode.

The estimate of the modal age value is 38 years which agrees with the value obtained using formula $37.9 \approx 38$ years.

Geometric mean: geometric mean is defined as the nth root of the product of n observations.

$x_1, x_2, \dots, x_n$ are set of n observations. The geometric mean of these observations is being denoted by $GM = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n}$ where x_i is the individual observations and n is the total number of the observations. The above formula is for ungrouped set of data.

Example 3.12

Below are scores of some accountancy students in management accounting test 2, 7, 5, 5, 8, 3 and 6. You are required to find GM of the students' scores.

Solution

Using the ungrouped formula set of data $GM = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n}$ where n is the sample (no. of observations) and x is one of the observations, we substitute our variables into the formula, thus:

$$GM = \sqrt[7]{2 \times 3 \times 5 \times 5 \times 6 \times 7 \times 8}$$

$$GM = \sqrt[7]{6 \times 5 \times 5 \times 6 \times 7 \times 8}$$

$$GM = \sqrt[7]{30 \times 5 \times 6 \times 7 \times 8}$$

$$GM = \sqrt[7]{150 \times 6 \times 7 \times 8}$$

$$GM = \sqrt[7]{900 \times 7 \times 8}$$

$$GM = \sqrt[7]{50400}$$

Now use scientific calculator and not market women calculator that makes the noise, pee, pee, while pressing the keys.

1st press 7 being its total no. of the observations.

2nd press your Shift Key.

3rd press root key which has sign like $\sqrt[n]{}$.

4th press the figures inside the root.

5th press equal to for the answer as $4.6965 \approx 4.7$.